Pivotal Tokens Are Linearly Predictable Before They Are Generated

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Abstract

Pivotal Token Search (PTS) labels a generated token as pivotal when it moves the model's probability of reaching a correct answer past a threshold, estimated by sampling many rollouts and adjudicating each with an oracle. We ask whether that expensive, outcome-grounded property is already present in the model's activations *one position before* the token is emitted — the only position at which a prediction could still gate an intervention. Using a reimplementation of the search that batches rollouts across queries and emits exact token positions, we generate events for Qwen3-0.6B over 3,000 GSM8K questions and train linear probes at $t-1$. The probe reaches **0.807 AUROC** and beats every cheaper explanation we tested, including a token-identity lookup table ($+0.110$, CI $[+0.050, +0.169]$) and next-token entropy ($+0.112$, CI $[+0.014, +0.213]$), where differences are bootstrapped over questions on paired resamples. The margin over uncertainty is real but modest, and we show it is highly sensitive to a design choice that is easy to leave implicit: drawing negative examples from the whole rollout rather than from the span where pivots occur inflates entropy from 0.693 to 0.814 and erases the probe's advantage entirely. Moving the probe to the pivot token itself raises it to 0.868 but inverts entropy to *below chance* (0.449), showing that predictive uncertainty peaks immediately before a pivotal token and collapses immediately after — the signature of a decision being resolved, and an argument for the $t-1$ framing on its own terms. We report a negative result on the sharper question of whether the *sign* of the impending shift is decodable (0.614, CI $[0.482, 0.735]$), and document two further ways this measurement inflates: collapsing a question's sampled rollouts to one canonical sequence discards 74% of the supervision, and selecting the best of 29 layers on the reporting split adds 0.063 AUROC.

1 Introduction

A language model solving a multi-step problem does not distribute its risk evenly across the tokens it emits. The Phi-4 technical report [Abdin et al., 2024] formalized this as *Pivotal Token Search* (PTS): a token t is pivotal when appending it to a prefix moves the estimated probability of eventually solving the task past a threshold,

$$|\Delta p| = \big|P(\text{success} \mid \text{prefix} + t) - P(\text{success} \mid \text{prefix})\big| > \tau. \tag{1}$$

Estimating either term requires sampling many completions and adjudicating each with an oracle, so PTS is an offline instrument that costs tens of rollouts per candidate position.

Our starting question was whether that instrument can be amortized: is pivotality linearly decodable from the residual stream at $t-1$, before the token is produced? A probe read one step early is a different object from one read at the pivotal token itself—the latter is a post-hoc classifier, the former is available while there is still a decision to influence.

It is. But the more informative result is *what else* predicts it. The question is only interesting if the answer is not already known for a cheaper reason, and two cheaper reasons exist. *Token identity*: some

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tokens may simply be pivotal more often, in which case a lookup table suffices and no representation is implicated. *Next-token uncertainty*: high-entropy "forking" tokens are the incumbent notion of a critical position [Wang et al., 2025].

The probe clears both, but the margin over uncertainty is small enough that how the comparison is set up decides the answer. We show that an unexamined choice about where negative examples are drawn from can erase it entirely, and we report the comparison both ways.

Our contributions:

- Evidence that pivotality is linearly decodable at $t-1$ (0.807 AUROC), above token identity, next-token entropy, top-2 margin, top-1 probability, and a random direction, using a paired bootstrap over questions (Section 5.1).
- A demonstration that this comparison is fragile: sampling negatives from the whole rollout rather than the pivot-bearing span raises entropy from 0.693 to 0.814 and removes the probe's advantage, because rollouts run to the token limit and their tails are fluent and low-entropy (Section 5.2).
- A contrast between reading at $t-1$ and at the pivot itself, in which entropy inverts from informative to below chance — evidence that uncertainty peaks before a pivotal token and collapses after it (Section 5.3).
- A negative result on whether the *sign* of the shift is decodable at this scale (Section 5.4), and a characterization of two ways the headline inflates: branch collapse and layer selection (Sections 3.1 and 5.5).

2 Related Work

Pivotal tokens. Abdin et al. [2024] introduce PTS to build preference pairs for Phi-4 post-training; Sharma [2025] released an open implementation and event datasets. Both use PTS as a data-generation step; the representational question—whether the property is present in activations beforehand—has not to our knowledge been asked, nor has the relationship between PTS labels and uncertainty been measured.

Uncertainty as the incumbent account. Wang et al. [2025] show that roughly 20% of tokens carry high entropy, act as branch points, and account for most of the gain from reinforcement learning with verifiable rewards. Sections 5.1 and 5.2 measure how much of PTS's criterion that account already covers, and how sensitive the measurement is.

Sentence-scale reasoning importance. Thought Anchors [Bogdan et al., 2025] identifies influential sentences by counterfactual resampling. We work at token scale, where the $t-1$ prediction problem is well posed: a sentence boundary is not a decoding step.

Probing and steering. Linear probes on intermediate representations [Alain and Bengio, 2018, Marks and Tegmark, 2023]; contrastive activation addition [Rimsky et al., 2024, Huang, 2024]; conditional steering [Lee et al., 2024]. Process reward models and value probes [Li et al., 2024] estimate $P(\text{success} \mid \text{prefix})$, a *level*; pivotality is $|\Delta p|$, a *sensitivity*.

3 Method

3.1 Generating the labels

We reimplement token-scale PTS with the same semantics as Sharma [2025] but express the bisection as a state machine, so that the rollouts of many questions can be batched together. The tree is sequential within a question—both children of a node need the node's own midpoint estimate—but questions are independent, and batching across them gives $8.9\times$ the generation throughput of running one node at a time. Emitted events carry the pivotal token's exact absolute position and the full searched rollout.

Two construction details materially affect the result.

Branches, not a canonical sequence. PTS samples several rollouts per question and they diverge after a few tokens. A construction that collapses a question to its longest rollout and matches the others against it as string prefixes silently discards every pivot on a diverging branch. On the released Qwen3-0.6B events, 104 questions carry 1,245 pivotal tokens over 447 distinct branches, and collapsing retains 318 of them—a 74% loss. We treat every maximal branch as its own sequence, and because branches share prefixes (and the residual stream at position p depends only on tokens $0 \dots p$) we label each distinct context exactly once.

Positions are exact, not inferred. Byte-pair encoding does not guarantee $\mathrm{enc}(a+b) = \mathrm{enc}(a)+\mathrm{enc}(b)$, so recovering a pivot's index by re-tokenizing its prefix is unsafe: a pivot token such as "." frequently merges with the preceding word. For data we generate this problem does not arise. For the released datasets, which predate the position field, we verify each recovered index against the reported token id and discard rows that fail.

3.2 Label noise in the oracle

$\widehat{\Delta p}$ is a difference of two binomial estimates from S rollouts each, accepted when $|\widehat{\Delta p}| \geq \tau$. Under the null of no true change $\mathrm{Var}(\widehat{\Delta p}) = 2p(1-p)/S$, so at the reference setting $S{=}50$, $\tau{=}0.2$ the acceptance threshold sits at only 2σ: a false-acceptance rate near 4.6% per tested position at $p{=}0.5$. We therefore store the raw counts $(n_{\text{before}}, s_{\text{before}}, n_{\text{after}}, s_{\text{after}})$ rather than a verdict, which makes τ a post-hoc knob and lets every event carry a Wilson interval. [TODO: report the accuracy gap between the full label set and a high-confidence subset.]

3.3 Probes, baselines, and how they are compared

Probes are ℓ_2-regularized logistic regressions on the residual stream, fit on standardized features and converted back to raw activation space so the resulting vector is usable as a direction. Layer and regularization strength are chosen on an inner split of the *training* questions; the test split is used once.

Baselines: token identity, as a smoothed log-odds lookup and as one-hot logistic regression, both fit on train and scored on held-out rows; next-token entropy, negative top-2 margin, and negative top-1 probability, computed from the model's own distribution at $t-1$; and a random unit direction.

Comparisons are paired. Confidence intervals are bootstrapped over *questions*, since positions within one reasoning trace are not independent. Crucially, when comparing the probe against a baseline we bootstrap the *difference* of AUROCs on the same resampled questions, rather than checking whether two marginal intervals overlap. The latter is badly conservative and would, on our data, have supported a conclusion the paired test rejects.

4 Experimental Setup

Model: Qwen3-0.6B [Qwen Team, 2025], 28 layers, $d{=}1024$. Task: GSM8K [Cobbe et al., 2021], conditioned on the bare question without a chat template, matching the released datasets and therefore without extended-thinking mode. Search: 3,000 candidate questions, $S{=}40$ rollouts per estimate, one searched rollout per question, $\tau{=}0.2$, 320 max new tokens, run on a single Blackwell-class GPU.

PTS only visits questions whose baseline success probability lies inside $[0.2, 0.8]$: a question the model always or never solves has no pivotal token, because a saturated probability cannot move. Of 3,000 candidates, 456 yielded at least one event, giving 688 pivotal positions and an equal number of sampled non-pivotal positions. Split by question: 365 train questions (1,132 rows) and 91 test questions (244 rows). Activations are read at all 29 residual-stream positions in bfloat16 at the labelled positions only.

Table 1: Predicting whether the *next* token is pivotal, from the residual stream at $t-1$. Qwen3-0.6B on GSM8K; negatives drawn from the pivot-bearing span. $P(\Delta \leq 0)$ is the fraction of question-level resamples in which the baseline matches or beats the probe.

Detector	AUROC	Probe $-$ baseline	95% CI	$P(\Delta \leq 0)$
Linear probe	**0.807**	—	—	—
Mean-difference direction	0.774	$+0.034$	$[+0.015, +0.055]$	0.001
Token identity (one-hot LR)	0.716	$+0.091$	$[+0.032, +0.154]$	0.001
Token identity (freq. table)	0.696	$+0.110$	$[+0.050, +0.169]$	< 0.001
Negative top-1 probability	0.704	$+0.101$	$[+0.003, +0.201]$	0.022
Negative top-2 margin	0.699	$+0.106$	$[+0.006, +0.207]$	0.021
Next-token entropy	0.693	$+0.112$	$[+0.014, +0.213]$	0.013
Random direction	0.478	$+0.328$	$[+0.222, +0.427]$	< 0.001

5 Results

5.1 Pivotality is decodable at $t-1$

At the selected layer (L=22) and penalty (C=10^{-4}) the probe reaches **0.807 AUROC** on 232 held-out positions from 91 unseen questions. Table 1 reports every detector together with the paired difference from the probe, bootstrapped on the same resampled questions.

Every difference has an interval excluding zero, so pivotality is present in the residual stream one step before the token that realizes it, and is explained neither by which token is about to be emitted nor by how uncertain the model is about it. The uncertainty margins are the smallest, at $+0.10$ to $+0.11$ with lower bounds near zero: on 91 questions we can separate them, but not comfortably.

5.2 Where the negatives come from decides the answer

That comparison depends on a choice which is easy to leave implicit. Our rollouts run to the token limit, and the natural thing to do is sample non-pivotal positions uniformly across the stored sequence. Doing so draws most of them from the fluent tail, *after* the model has committed to an answer: 63% of negatives landed beyond the last pivot, with mean next-token entropy 0.275 against 0.476 before it. Median position was 232 for negatives against 146 for pivots, so raw position became informative too.

Under that sampling, entropy rises from 0.693 to 0.814 and the probe's advantage over it falls to $+0.039$ with a confidence interval spanning zero — a null result manufactured entirely by the negative distribution. We therefore restrict negatives to the span in which pivots occur, and we recommend that any comparison of this kind state where its negatives came from. Published PTS datasets avoid the problem incidentally, because the sequence they store stops at the deepest pivot; it appears as soon as one keeps the full rollout, which is otherwise the more useful artifact.

5.3 Reading at the pivot instead of before it

Moving the label from $t-1$ to the pivot itself gives a post-hoc classifier: informative, but unable to gate anything, because by then the token exists. Table 2 compares the two.

The probe gains six points and token identity nine, which is unsurprising: at t the token occupying the labelled position *is* the pivot. The informative change is entropy, which inverts from 0.693 to 0.449 — below chance. Uncertainty is elevated immediately before a pivotal token and suppressed immediately after it, which is what a decision point being resolved should look like, and it means the two positions carry opposite uncertainty signatures rather than differing in degree. It also supplies an argument for the $t-1$ framing that does not depend on the intervention story: $t-1$ is where the uncertainty signal actually lives.

Table 2: Same data and same negatives, with the probe read at $t-1$ versus at the pivotal token t.

Detector	at $t-1$	at t
Linear probe	0.807	**0.868**
Mean-difference direction	0.774	0.810
Token identity (freq. table)	0.696	0.786
Next-token entropy	**0.693**	**0.449**
Negative top-2 margin	0.699	0.445
Random direction	0.478	0.514

5.4 The sign is not decodable at this scale

Pivotality as defined in Eq. 1 is symmetric: a token that rescues an answer and one that ruins it receive the same label. The sharper question is whether the *sign* is decodable at $t-1$. Entropy is sign-blind by construction and cannot answer it, so a positive result here would be immune to Section 5.2 entirely.

Restricting to pivotal positions and relabelling by $\mathrm{sign}(\Delta p)$ gives 566 training and 122 test rows (56.5% helpful). A probe reaches 0.614 AUROC with a 95% interval of $[0.482, 0.735]$ that contains chance; the signed mean-difference direction reaches 0.567 and token identity 0.555. We report this as a negative result. An earlier version on a smaller, noisier label set gave 0.717 and looked promising; it did not survive four times as many held-out questions.

5.5 No layer is special, and selection inflates the headline

Per-layer accuracy spans 0.651–0.741 with mean 0.702. Simulating a null in which all 29 layers are equally informative, each measured on 232 rows, gives an expected maximum of 0.762 against an observed 0.741, with $P(\text{observed or higher}) = 0.96$. There is no evidence of layer specialization; the signal is spread across the mid-to-late stack, and naming a single "pivotal layer" would be an artifact of maximizing over correlated noise.

The cost of getting the protocol wrong is measurable. Choosing the layer on training data and reporting on test gives 0.807; taking the best layer on the reporting split gives 0.814, a gap of 0.007. Under a leakier protocol we measured the same gap at 0.063 — large enough to manufacture a result.

5.6 Regularization

AUROC moves from 0.752 at $C{=}1$ to 0.816 at $C{=}10^{-3}$, and the probe direction rotates toward the mean-difference direction as the penalty tightens (cosine 0.173 to 0.765). A weakly regularized probe still classifies, but along a substantially different direction — which matters for anyone intending to use the weights as an intervention direction rather than only as a classifier.

6 Limitations

A single 0.6B model on a single task family, with 91 held-out questions. The oracle's labels are themselves noisy (Section 3.2) in a way we have quantified analytically but not yet empirically. The acceptance filter retains only mid-difficulty questions, so pivot density is confounded with task difficulty relative to model capability and this label set is not a uniform sample of GSM8K. Conditioning on the bare question puts an instruct model off distribution; we observed rollouts in which the model hallucinated multiple-choice options, and at least one event whose extreme Δp traces to the answer-extraction cascade matching such a hallucination rather than to a genuine shift. Linear probes test linear decodability only. Everything here is correlational: we have not shown that the direction the probe identifies is causally used.

Most importantly, the margins over uncertainty in Table 1 are $+0.10$ to $+0.11$ with lower bounds within 0.01 of zero, on 91 questions. They are separations, not comfortable ones, and Section 5.2 shows how little it takes to erase them. [TODO: Scale the question count and re-test; extend to Qwen3-1.7B and 4B.]

7 Conclusion

Pivotality is linearly present in a model's residual stream one position before the token that realizes it, at 0.807 AUROC, above token identity and above every next-token uncertainty statistic we tested. The margin over uncertainty is modest and, we show, contingent on where negative examples are drawn from: a plausible-looking choice removes it entirely. Reading at the pivot instead raises the probe but inverts entropy to below chance, locating the uncertainty signal specifically at $t-1$ — the one position that is both informative and available while a decision can still be influenced. Whether the *direction* of the impending shift is also decodable remains open; at this sample size it is not.

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A Reproduction

Events: `stvngo/Qwen3-0.6B-pts-tokens`. Activations: `stvngo/Qwen3-0.6B-pivotal-activations`. All numbers in Section 5 come from `artifacts/probes_v2_full/qwen3-0.6b_report.json`, produced by `scripts/pts_run.py`, `scripts/build_and_extract.py` and `scripts/evaluate_probes_v2.py`. [TODO: Add the layer sweep and regularization-path figures.]